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Further Mathematics for CCEA A-Level: Statistics

Complete AS and A2 Course

CCEA GCE Further Mathematics • ISBN 978-1-0666184-8-4

by Isaac Teng — MSci Mathematics & Physics, UCL; Founder, IREducation Inc.

What this evaluation copy contains

These are complete, unaltered excerpts from the full textbook. Published page and chapter numbers are retained so that the sample can be compared directly with the complete book.

- **Front matter** — Introduction, About the Author, and the full Contents.
- **Chapter 4** — Linear Combinations, Sampling and Estimation (pp. 67–90), with solutions to Exercises 4A–4F. End-of-chapter mixed-exercise solutions are reserved for the complete book.
- **Appendix A excerpt** — calculator guide Sections A.1–A.4: random integers, one-variable statistics, correlation and regression, and probability distributions.
- **A21 Practice Set** — the first A2 Statistics practice set, with its matching solutions.

The full book at a glance

- 192 pages and five chapters covering the complete CCEA Statistics option at AS and A2.
- Six practice sets — three at AS and three at A2 — with complete worked solutions.
- A complete calculator guide, statistical reference and per-chapter worked answers.

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Further Mathematics for CCEA A-Level: Statistics

Complete AS and A2 Course

Isaac Teng Chit Hang

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A2 Statistics assumes the AS Statistics section as well as the compulsory pure units.

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Introduction

This textbook is the **Statistics** volume in the wider IRMaths series for **CCEA GCE Further Mathematics**. Companion volumes cover the compulsory Pure Mathematics units and the other applied options. This volume provides complete coverage of the Statistics option at AS and A2 and is intended to be used alongside the relevant Pure Mathematics material.

The five chapters follow the CCEA Statistics pathway. Chapters 1–3 cover the AS material: sampling and combinatorial probability, probability distributions, and correlation and regression. Chapters 4–5 develop the A2 material: linear combinations, sampling distributions, estimation, confidence intervals, Student's t procedures and χ^2 inference. They assume familiarity with the statistical language, probability models and algebra developed at AS.

The aim is to guide students from basic statistical reasoning to confident work with unfamiliar problems. Statistics is not simply a collection of calculator procedures: a complete solution must select an appropriate model, recognise its assumptions, carry out the mathematics accurately and interpret the result in context. The text therefore emphasises precise notation, justified method selection and clear conclusions as well as numerical calculation.

Each chapter begins with its learning goals and develops the ideas through explanation, key results and worked examples. Section exercises provide focused practice; the end-of-chapter mixed exercises bring methods together and progress from routine fluency to multi-step modelling, interpretation and the diagnosis of common errors. Fully worked answers allow students to check both their result and the structure of their reasoning.

The six practice sets—three at AS and three at A2—provide timed consolidation once the relevant content is secure. They should be used alongside the latest official specification, specimen materials, past papers and instructions to candidates.

The calculator guide gives model-specific key sequences for the Casio fx-991EX, fx-991CW and fx-CG50. It supports the mathematics rather than replacing it: hypotheses, distributions, substitutions, degrees of freedom and conclusions should still be shown. The Statistical Reference appendix gathers the principal notation and formulae used throughout the course.

The book can be used in class, for independent study or as a revision reference, and alongside the IRMaths platform (`irmaths.ireducation.org`), which provides further resources across the CCEA Further Mathematics series.

Specifications, assessment arrangements and calculator software can change, so students and teachers should consult current CCEA documents. Corrections and suggestions are welcome at `isaacteng@ireducation.org`; please identify the edition, page and exercise number.

Thank you for choosing this book. I hope that it helps you develop not only the techniques required for the examination, but also the judgement and clarity that make statistics useful beyond it.

A Note on Beyond the Syllabus Content

Purple Beyond the Syllabus boxes appear only where they illuminate an assessed idea or prepare for later mathematics. They are optional and are not required for the CCEA examination.

About the Author

Isaac Teng is the founder of IREducation Inc., an ed-tech organization based in Hong Kong SAR. He is also an MSci Mathematics and Physics student at University College London (UCL), and the lead developer for the IREducation platform, including IRMaths.

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Chapter 4

Linear Combinations, Sampling and Estimation

After completing this chapter you should be able to:

- Use expectation and variance for linear combinations of independent random variables.
- Solve problems involving independent normal variables, sums of observations and scalar multiples.
- Apply the central limit theorem for samples of at least 30.
- Calculate point estimates, unbiased variance, standard error and confidence intervals for a mean.

Chapter route

This chapter connects probability models to statistical estimation. It begins by finding the mean and variance of totals, differences and other linear combinations. Those rules describe sampling distributions; the central limit theorem then explains why many sample means are approximately normal. Finally, a point estimate is paired with its standard error and a critical value to form a confidence interval. Keep the chain

$$\text{random variables} \longrightarrow \text{sampling distribution} \longrightarrow \text{standard error} \longrightarrow \text{interval estimate}$$

in view: each stage supplies the justification for the next.

4.1 Linear Combinations of Random Variables

Often in statistics, we need to combine random variables. For example, finding the total weight of items in a box, or the difference in scores between two competitors.

4.1.1 Expectation and Variance Rules

If X and Y are random variables, and a and b are constants, we can find the expectation (mean) and variance of a linear combination $aX \pm bY$.

Key Point

For any random variables X and Y, and constants a and b:

$$E(aX \pm bY) = aE(X) \pm bE(Y)$$

For **independent** random variables X and Y:

$$\text{Var}(aX \pm bY) = a^2 \,\text{Var}(X) + b^2 \,\text{Var}(Y)$$

Common Misconception

For independent random variables, a common error is subtracting variances. **The variance contributions add even when the variables are subtracted.**

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$$

Without independence, a covariance term is required; never write $\text{Var}(X) - \text{Var}(Y)$.

4.1.2 Multiples vs. Sums of Independent Observations

You must carefully distinguish between multiplying a single observation by a constant, and adding together multiple independent observations from the same distribution.

- **Multiple of a single observation (aX):** Imagine pouring a cup of water (X) and then multiplying its volume by 3.

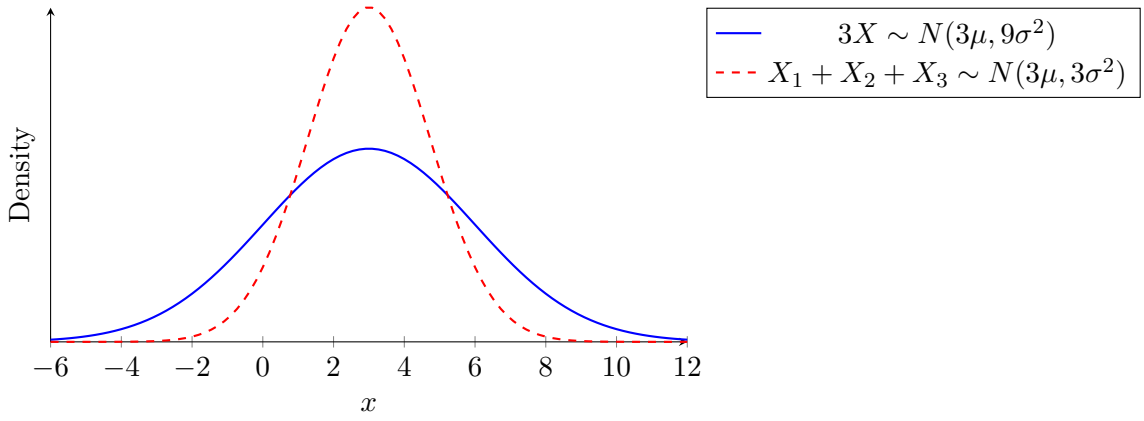
$$E(3X) = 3E(X)$$
$$\text{Var}(3X) = 3^2 \,\text{Var}(X) = 9 \,\text{Var}(X)$$

- **Sum of independent observations ($X_1 + X_2 + X_3$):** Imagine pouring three separate, independent cups of water.

$$E(X_1 + X_2 + X_3) = E(X) + E(X) + E(X) = 3E(X)$$
$$\text{Var}(X_1 + X_2 + X_3) = \text{Var}(X) + \text{Var}(X) + \text{Var}(X) = 3 \,\text{Var}(X)$$

Notice that while the expectations are the same, the variances are very different. The sum of independent observations has a much smaller variance than a multiple of a single observation, because independent random variations tend to cancel each other out.

The diagram below illustrates this difference for a normally distributed random variable $X \sim N(\mu, \sigma^2)$, comparing the distributions of $3X$ and $X_1 + X_2 + X_3$.



Example 2

Question: The random variable X has mean 10 and variance 4. The random variable Y has mean 15 and variance 5. X and Y are independent. Find:

(a) $E(3X - 2Y)$

(b) $\text{Var}(3X - 2Y)$

(c) $E(X_1 + X_2 + X_3 + X_4)$

(d) $\text{Var}(X_1 + X_2 + X_3 + X_4)$

Solution:

(a) $E(3X - 2Y) = 3E(X) - 2E(Y) = 3(10) - 2(15) = 30 - 30 = 0.$

(b) $\text{Var}(3X - 2Y) = 3^2\,\text{Var}(X) + (-2)^2\,\text{Var}(Y) = 9(4) + 4(5) = 36 + 20 = 56.$

(c) $E(X_1 + X_2 + X_3 + X_4) = 4E(X) = 4(10) = 40.$

(d) $\text{Var}(X_1 + X_2 + X_3 + X_4) = 4\,\text{Var}(X) = 4(4) = 16.$

Exercise 4A

1 The independent random variables U and V satisfy $\text{E}(U) = 8$, $\text{Var}(U) = 3$, $\text{E}(V) = 12$ and $\text{Var}(V) = 5$.

(a) Find $\text{E}(4U - V + 7)$. **[2 marks]**

(b) Find $\text{Var}(4U - V + 7)$. **[2 marks]**

(c) Find $\text{E}(V^2)$. **[1 mark]**

2 In a precision assembly process, the thickness of a ceramic spacer, C mm, has mean 4.2 mm and standard deviation 0.15 mm. The thickness of a steel washer, W mm, has mean 1.8 mm and standard deviation 0.08 mm. All components are produced independently.

(a) Find the mean and variance of the total thickness of a stack made from 4 randomly selected ceramic spacers and 3 randomly selected steel washers. **[3 marks]**

(b) Find the mean and standard deviation of the clearance gap $G = C - 2W$, where one ceramic spacer is compared against twice the thickness of a single steel washer. **[2 marks]**

3 A forestry drone carries rechargeable battery packs. The energy capacity of a randomly chosen pack, B Wh, has mean 180 Wh and variance 16 Wh^2.

(a) Find the mean and variance of $B_1 + B_2 + B_3$, where B_1, B_2, B_3 are three independent battery packs, and compare these with the mean and variance of $3B$. **[3 marks]**

(b) A technician models the total energy of three independent battery packs fitted to the drone as $3B$. Explain what physical situation $3B$ actually represents and why $B_1 + B_2 + B_3$ has a smaller variance. **[2 marks]**

4 A ferry operator charges £35 per car and £120 per commercial van on a morning crossing. On a given weekday morning, the number of cars, X, has mean 42 and standard deviation 6, and the number of commercial vans, Y, has mean 15 and standard deviation 4. Assume Initially that X and Y are independent.

(a) Find the mean and standard deviation of the total vehicle revenue, $R = 35X + 120Y$, on a weekday morning. **[3 marks]**

(b) Explain why the assumption that X and Y are independent might be questionable in practice, and state which part of your calculation in part (a) relies on this assumption. **[2 marks]**

5 The independent random variables P and Q satisfy $\mathrm{E}(P) = 6$, $\mathrm{Var}(P) = \sigma^2$, $\mathrm{E}(Q) = 9$ and $\mathrm{Var}(Q) = 2\sigma^2$. The random variable W is defined by $W = kP - 2Q$, where k is a positive constant.

(a) Given that $\mathrm{E}(W) = 12$, show that $k = 5$. **[2 marks]**

(b) Given also that the standard deviation of W is $11\sqrt{3}$, find the value of σ^2. **[3 marks]**

(c) A student claims that $\mathrm{Var}(P_1 + P_2 - P_3) = \mathrm{Var}(P_1) + \mathrm{Var}(P_2) - \mathrm{Var}(P_3) = \sigma^2$ for three independent observations of P. Correct the student's working and state the true value of $\mathrm{Var}(P_1 + P_2 - P_3)$. **[1 mark]**

6 ⋆ Two independent instruments, Instrument 1 and Instrument 2, are used to measure the concentration μ of a dissolved mineral in water. A single reading M_1 from Instrument 1 has mean μ and variance 9 $\mathrm{mg^2L^{-2}}$, while a single reading M_2 from Instrument 2 has mean μ and variance 16 $\mathrm{mg^2L^{-2}}$. A combined reading is formed using the linear combination $C = wM_1 + (1 - w)M_2$, where $0 < w < 1$.

(a) Show that $\mathrm{E}(C) = \mu$ for all values of w. **[2 marks]**

(b) Show that $\mathrm{Var}(C) = (25w^2 - 32w + 16)$ $\mathrm{mg^2L^{-2}}$. **[2 marks]**

(c) Find the value of w that minimises $\mathrm{Var}(C)$, and calculate this minimum variance. **[2 marks]**

4.1.3 Applications to Normal Distributions

A very powerful property of the normal distribution is that any linear combination of independent normal random variables is also normally distributed.



Figure 4.1: Carl Friedrich Gauss (1777–1855); the normal distribution is often called the Gaussian distribution in his honour.

Key Point

If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ are independent random variables, then the linear combination $aX \pm bY$ is also normally distributed:

$$aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$$

This allows us to calculate probabilities for complex practical situations, such as the total weight of packed items or comparing the scores of two competitors.

Packing and Total Capacity Problems

In these problems, you are typically asked to find the probability that the sum of several independent items exceeds a certain limit.

Example 3

Question: The mass of a randomly chosen apple, A, is normally distributed with mean 150 g and standard deviation 12 g. The mass of a randomly chosen pear, P, is normally distributed with mean 120 g and standard deviation 10 g. A bag contains 4 apples and 3 pears. Assuming the masses are independent, find the probability that the total mass of the fruit in the bag exceeds 1000 g.
Solution: Let T be the total mass. $T = A_1 + A_2 + A_3 + A_4 + P_1 + P_2 + P_3$
First, find the expectation of T:

$$E(T) = 4E(A) + 3E(P) = 4(150) + 3(120) = 600 + 360 = 960 \text{ g}$$

Next, find the variance of T. Since we are adding independent items, we add their variances:

$$\text{Var}(T) = 4\,\text{Var}(A) + 3\,\text{Var}(P) = 4(12^2) + 3(10^2) = 4(144) + 3(100) = 576 + 300 = 876$$

So, $T \sim N(960, 876)$.
We need to find $P(T > 1000)$. Using the normal cumulative distribution function on a calculator with $\mu = 960$, $\sigma = \sqrt{876} \approx 29.597$, Lower $= 1000$, Upper $= 99999$:

$$P(T > 1000) = 0.0883 \text{ (to 3 s.f.)}$$

Comparing Two Variables ($X > Y$)

A common exam question asks for the probability that one random variable is greater than another. To solve this, you must rearrange the inequality to compare a new combined variable to zero.

Key Point

To find $P(X > Y)$, rearrange it to $P(X - Y > 0)$. Define a new variable $W = X - Y$. Find the mean and variance of W, and then calculate $P(W > 0)$.

Example 4

Question: The time taken by Alice to run a 5km race, A, is normally distributed with mean 24 minutes and standard deviation 2 minutes. The time taken by Bob, B, is normally distributed with mean 25 minutes and standard deviation 1.5 minutes. Assuming their times are independent, find the probability that Alice beats Bob in a randomly chosen race.
Solution: Alice beats Bob if her time is less than Bob's time. We need to find $P(A < B)$. Rearrange this to: $P(A - B < 0)$.
Let $W = A - B$. Find the mean of W:

$$E(W) = E(A) - E(B) = 24 - 25 = -1$$

Find the variance of W. Remember, variances always add!

$$\text{Var}(W) = \text{Var}(A) + \text{Var}(B) = 2^2 + 1.5^2 = 4 + 2.25 = 6.25$$

So, $W \sim N(-1, 6.25)$.
We need to find $P(W < 0)$.
Using a calculator with $\mu = -1$, $\sigma = \sqrt{6.25} = 2.5$, Lower = -99999, Upper = 0:

$$P(W < 0) = 0.6554 \text{ (to 4 d.p.)}$$

There is a 65.5% chance that Alice beats Bob.

Exercise 4B

1 The independent random variables X and Y follow normal distributions with $X \sim \text{N}(32, 6^2)$ and $Y \sim \text{N}(20, 8^2)$.

(a) State the exact distribution of $X + Y$ and find $\text{P}(X + Y < 60)$. **[3 marks]**

(b) Find $\text{P}(2X - 3Y > 15)$. **[3 marks]**

2 An airline catering trolley is loaded with meal trays and beverage canisters. The mass of an empty trolley, T kg, is normally distributed with mean 28 kg and standard deviation 1.2 kg. The mass of a prepared meal tray, M kg, is normally distributed with mean 0.65 kg and standard deviation 0.06 kg. The mass of a filled beverage canister, B kg, is normally distributed with mean 4.5 kg and standard deviation 0.25 kg. All masses are independent. A fully stocked trolley consists of 1 empty trolley, 40 meal trays and 4 beverage canisters.

(a) Find the mean and variance of the total mass of a fully stocked trolley. **[3 marks]**

(b) For cabin crew safety, a loaded trolley should not exceed 75 kg. Find the probability that a randomly selected fully stocked trolley exceeds this safety limit. **[2 marks]**

3 The volume of solvent dispensed by an automated pipette in a single cycle, V ml, follows a normal distribution with mean 25.0 ml and variance 0.36 ml^2.

(a) In Method A, a chemist runs 4 independent cycles of the pipette into a flask. Find the probability that the total volume dispensed lies between 98.5 ml and 101.5 ml. **[3 marks]**

(b) In Method B, the pipette dispenses a single cycle and a scaling mechanism delivers exactly 4 times that single volume, $4V$. Find $\text{P}(98.5 < 4V < 101.5)$ and explain why this probability is smaller than in part (a). **[3 marks]**

4 In a biathlon relay stage, the time taken by Skier P, P minutes, is normally distributed with mean 34.2 and standard deviation 1.6. The time taken by Skier Q, Q minutes, is normally distributed with mean 35.0 and standard deviation 1.2. Assume that their times are independent.

(a) Find the probability that Skier P completes the stage in a faster time than Skier Q. **[3 marks]**

(b) Find the probability that Skier P finishes more than 2.0 minutes ahead of Skier Q. **[3 marks]**

5 A joinery workshop manufactures wooden dowels and drilled sockets. The diameter of a dowel, D mm, is normally distributed with mean 11.85 mm and standard deviation 0.12 mm. The internal diameter of a socket, S mm, is normally distributed with mean 12.10 mm and standard deviation 0.09 mm. All dowels and sockets are produced independently.

(a) For a joint to assemble properly without being loose, the socket diameter must exceed the dowel diameter by no more than 0.40 mm, so that $0 < S - D \leq 0.40$. Find the probability that a randomly chosen dowel and socket form a proper joint. **[4 marks]**

(b) Find the probability that 3 randomly chosen dowels placed side by side have a combined width greater than the combined internal diameter of 3 randomly chosen sockets. **[2 marks]**

6 ⋆ A pharmaceutical tablet consists of an active core and a protective coating. The mass of the core, A mg, follows $\mathrm{N}(240, 5^2)$, and the mass of the coating, B mg, follows $\mathrm{N}(\mu, \sigma^2)$, independently of A.

(a) Given that the mean total mass of a coated tablet is 275 mg, write down the value of μ. **[1 mark]**

(b) Quality control requires that the probability of a coated tablet having a total mass greater than 288 mg is 0.025. Find the value of σ^2, giving your answer to 2 decimal places. **[5 marks]**

4.2 Parameters, Statistics and Estimators

When conducting statistical investigations, it is rarely possible to measure every single item in a population. Instead, we take a **sample** and use it to make inferences about the whole **population**. To do this accurately, you must be precise with your terminology.

4.2.1 Parameters and Statistics

Key Point

- A **parameter** is a numerical characteristic of the entire population. It is usually a fixed, but unknown, constant. Examples include the population mean (μ) and population variance (σ^2).
- A **statistic** is a numerical characteristic calculated entirely from sample data. It must not contain any unknown parameters. Examples include the sample mean ($\bar{X}$) and sample variance (S^2).

Because a statistic is calculated from a random sample, the statistic itself is a random variable. If you took a different sample, you would likely get a different value for your statistic. The probability distribution of all possible values of a statistic is called its **sampling distribution**.

4.2.2 Unbiased Estimators and Standard Error

We use statistics to estimate the unknown parameters of a population. For example, we use the sample mean $\bar{X}$ to estimate the population mean μ.

Key Point

- An **unbiased estimator** is a statistic whose expected value (mean) is exactly equal to the population parameter it is estimating.

$$E(\text{Estimator}) = \text{Parameter}$$

  For example, the sample mean is an unbiased estimator of the population mean because $E(\bar{X}) = \mu$.

- The **standard error** is the standard deviation of the sampling distribution of a statistic. It measures how much the statistic is expected to vary from sample to sample.

For the sample mean $\bar{X}$ drawn from a population with variance σ^2, the variance of $\bar{X}$ is $\frac{\sigma^2}{n}$. Therefore, the standard error of the mean is:

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}}$$

Example 1

Question: A factory produces lightbulbs. The lifespan of the lightbulbs has an unknown population mean μ and a known population standard deviation of 120 hours. A quality control manager takes a random sample of 50 lightbulbs and calculates the sample mean lifespan to be 1050 hours.

(a) State the value of the parameter being estimated.

(b) State the value of the statistic used as an estimate.

(c) Calculate the standard error of the mean.

Solution:

(a) The parameter is the population mean, μ. Its exact value is unknown.

(b) The statistic is the sample mean, $\bar{x} = 1050$ hours.

(c) The standard error is $\frac{\sigma}{\sqrt{n}} = \frac{120}{\sqrt{50}} = \frac{120}{7.071} \approx 16.97$ hours.

Exercise 4C

1 An acoustic engineer measures the reverberation time, x seconds, at 6 randomly selected positions inside a concert hall:

$$1.8, \quad 2.1, \quad 1.9, \quad 2.4, \quad 2.0, \quad 1.8$$

(a) State whether the true mean reverberation time μ of the entire concert hall is a parameter or a statistic, giving a reason for your answer. **[1 mark]**

(b) Calculate unbiased point estimates of the population mean μ and the population variance σ^2. **[3 marks]**

(c) Calculate the estimated standard error of the sample mean. **[2 marks]**

2 To monitor dissolved oxygen in an estuary, a marine chemist takes a random sample of 16 water specimens. The concentrations, y mg L^{-1}, are summarised by:

$$\sum y = 136.0, \qquad \sum y^2 = 1174.0$$

(a) Calculate the unbiased point estimate of the population mean and the unbiased point estimate, s^2, of the population variance. **[3 marks]**

(b) A trainee calculates the variance estimate by dividing $\sum(y - \bar{y})^2$ by 16 to obtain 1.125. Explain why dividing by n produces a biased estimator of σ^2, and justify the divisor $n - 1$ using the concept of degrees of freedom. **[2 marks]**

3 Let X_1, X_2, X_3 be a random sample of 3 independent observations from a population with mean μ and variance σ^2. Consider the two statistics:

$$T_1 = \frac{X_1 + 3X_2 + X_3}{5}, \qquad T_2 = \frac{2X_1 + X_2}{3}$$

(a) Show that both T_1 and T_2 are unbiased estimators of μ. **[2 marks]**

(b) Find $\text{Var}(T_1)$ and $\text{Var}(T_2)$ in terms of σ^2, and state with a reason which of T_1 or T_2 is the more efficient estimator. **[3 marks]**

(c) Explain why the sample mean $\bar{X} = \frac{X_1+X_2+X_3}{3}$ is preferable to both T_1 and T_2 as an estimator of μ. **[1 mark]**

4 ⋆ A random sample of 3 independent observations Y_1, Y_2, Y_3 is drawn from a population with mean μ and variance σ^2. A statistic is defined as

$$W = aY_1 + bY_2 + \frac{1}{2}Y_3$$

where a and b are constants.

(a) Given that W is an unbiased estimator of μ, find an expression for b in terms of a. **[2 marks]**

(b) Given further that $\text{Var}(W) = \frac{3}{8}\sigma^2$, find the value of a and the value of b. **[3 marks]**

5 The tensile strength of carbon-fibre rods has population mean μ and population standard deviation $\sigma = 36$ MPa.

(a) Calculate the standard error of the sample mean for a random sample of size $n = 18$, and find the sample size required to reduce this standard error to 3 MPa. **[3 marks]**

(b) An engineer states: "Taking a larger sample of rods will reduce the population standard deviation σ, and tripling the sample size will divide the standard error by 3." Identify and correct both statistical misconceptions in this statement. **[2 marks]**

6 Two technicians independently test the pH of soil cores from the same field. Technician A takes a random sample of 5 cores and obtains sample mean $\bar{x}_A = 6.4$ and unbiased sample variance $s_A^2 = 0.15$. Technician B takes an independent random sample of 6 cores from the same field and obtains sample mean $\bar{x}_B = 6.4$ and unbiased sample variance $s_B^2 = 0.25$.

(a) Explain briefly why the combined sample of 11 cores also has sample mean $\bar{x} = 6.4$. **[1 mark]**

(b) Calculate the unbiased point estimate s^2 of the population variance using all 11 cores combined, and hence estimate the standard error of the combined sample mean. **[4 marks]**

Beyond the Syllabus: Bessel's Correction: Why do we divide by $n-1$?

When calculating the variance of a whole population, we simply find the average of the squared deviations from the mean by dividing by the population size, N. So why, when calculating the sample variance (s^2), do we suddenly divide by $n-1$?
This adjustment is known as **Bessel's Correction**, and it solves a fundamental problem in statistics: **underestimation bias**.



Figure 4.2: Friedrich Wilhelm Bessel (1784–1846)

1. The Problem of Underestimation
When we calculate the variance of a sample, we are trying to estimate the spread of the entire population. To do this, we measure how far each data point is from the mean.
However, we rarely know the true population mean (μ). Instead, we have to use the sample mean ($\bar{x}$) as a substitute. Here is the mathematical catch: by definition, a sample of data will always be clustered more closely around its *own* sample mean than it will be around the true population mean.
Because $\sum(x-\bar{x})^2$ is mathematically guaranteed to be smaller than or equal to $\sum(x-\mu)^2$, dividing by n would *underestimate* the true population variance on average. To fix this, we divide by a slightly smaller number $(n-1)$, which inflates the result just enough to give an **unbiased estimate**.
2. Degrees of Freedom
But why exactly $n-1$ and not $n-2$ or $n-1.5$? This brings in the concept of **Degrees of Freedom**.
Imagine you have a sample of $n=3$ numbers, and you calculate their sample mean to be 10. Suppose the first two numbers in your sample are 8 and 12. What *must* the third number be? For the mean of three numbers to be 10, their sum must be 30. Since $8+12=20$, the third number has absolutely no choice: it *must* be 10.
Because we forced the data to match our calculated sample mean ($\bar{x}$), the last data point has no "freedom" to vary. It is mathematically locked in. In general, once you calculate a sample mean from n data points, you only have $n-1$ independent pieces of information left to calculate the variance. Therefore, we average the variance over the $n-1$ degrees of freedom, rather than n. Indeed, it can be shown that $E\left[\sum(x-\bar{x})^2\right]=(n-1)\sigma^2$, so dividing by $n-1$ makes $E(s^2)=\sigma^2$ exactly.

4.3 The Central Limit Theorem

If a population is normally distributed, $X \sim N(\mu, \sigma^2)$, then the distribution of the sample mean $\bar{X}$ is exactly normally distributed for any sample size n:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

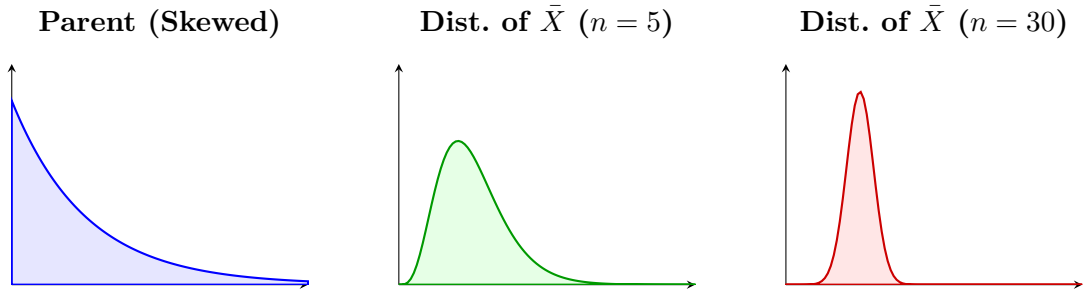
However, in the real world, many populations are not normally distributed. They might be skewed,

uniform, or completely unknown. This is where the **Central Limit Theorem (CLT)** becomes one of the most powerful tools in statistics.

Key Point

The Central Limit Theorem (CLT) states that if you take a sufficiently large random sample ($n \geq 30$) from *any* population with mean μ and variance σ^2, the sampling distribution of the sample mean $\bar{X}$ is approximately normally distributed:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$



Key Point

Choosing and justifying the sampling distribution

- If the population is normal, $\bar{X}$ is normal for every sample size; the Central Limit Theorem is not needed.
- If the population is non-normal or its distribution is unknown, use the Central Limit Theorem when the condition $n \geq 30$ is satisfied.
- The variance of the sample mean is σ^2/n, so its standard error is $\sigma/\sqrt{n}$.
- When justification is requested, name the condition that makes the normal model valid.

4.3.1 Applying the Central Limit Theorem

When calculating probabilities for a sample mean from a non-normal distribution, you will first define the approximate normal distribution using the CLT, and then use your calculator's normal cumulative distribution function.

Example 3

Question: The time taken, X minutes, for a mechanic to service a car has an unknown distribution with mean $\mu = 45$ minutes and standard deviation $\sigma = 8$ minutes. A random sample of 40 car services is monitored.

(a) Find the probability that the mean time taken for these 40 services is less than 43 minutes.

(b) State, with a reason, whether you have used the Central Limit Theorem in your calculation.

Solution:

(a) We are given $\mu = 45$, $\sigma = 8$, and $n = 40$. The variance of the population is $\sigma^2 = 8^2 = 64$.

Since $n = 40 \geq 30$, we can apply the Central Limit Theorem to approximate the distribution of the sample mean $\bar{X}$:

$$\bar{X} \approx N\left(45, \frac{64}{40}\right)$$

$$\bar{X} \approx N(45, 1.6)$$

We want to find $P(\bar{X} < 43)$. Using a calculator with lower bound -9999, upper bound 43, $\mu = 45$, and $\sigma = \sqrt{1.6} \approx 1.2649$:

$$P(\bar{X} < 43) = 0.0569 \text{ (to 4 d.p.)}$$

(b) Yes, the Central Limit Theorem was used. It was necessary because the underlying population distribution of service times was unknown, but the sample size was large ($n = 40 \geq 30$), allowing us to assume $\bar{X}$ is approximately normally distributed.

Example 4

Question: The continuous random variable Y has a uniform distribution over the interval $[10, 30]$. A random sample of 50 observations of Y is taken. Find the probability that the sample mean is greater than 21.

Solution: First, find the mean and variance of the uniform distribution $Y \sim U(10, 30)$.

$$\mu = \frac{a+b}{2} = \frac{10+30}{2} = 20$$

$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{(30-10)^2}{12} = \frac{400}{12} = \frac{100}{3}$$

We are taking a sample of $n = 50$. Since $n \geq 30$, by the Central Limit Theorem, the sample mean $\bar{Y}$ is approximately normally distributed:

$$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\bar{Y} \approx N\left(20, \frac{100/3}{50}\right)$$

$$\bar{Y} \approx N\left(20, \frac{2}{3}\right)$$

We need to find $P(\bar{Y} > 21)$. Using a calculator with lower bound 21, upper bound 9999, $\mu = 20$, and $\sigma = \sqrt{\frac{2}{3}} \approx 0.8165$:

$$P(\bar{Y} > 21) = 0.1103 \text{ (to 4 d.p.)}$$

4.3.2 Working Backwards: Finding Sample Size

Sometimes you will be given a target probability or a target standard error and asked to find the minimum sample size required.

Example 5

Question: The mass of a certain type of rock is known to have a standard deviation of 4.5 kg. A geologist wants to take a large random sample of rocks such that the standard error of the mean is less than 0.5 kg. Find the minimum sample size the geologist must take.
Solution: The standard error of the mean is given by $\frac{\sigma}{\sqrt{n}}$. We are given $\sigma = 4.5$ and we require:

$$\frac{4.5}{\sqrt{n}} < 0.5$$

Multiply both sides by $\sqrt{n}$ (since $n > 0$, the inequality sign does not change):

$$4.5 < 0.5\sqrt{n}$$

Divide by 0.5:

$$9 < \sqrt{n}$$

Square both sides:

$$81 < n$$

Since n must be an integer, the minimum sample size is $n = 82$.

Exercise 4D

1 The waiting time, T seconds, for a database server to process an archival query has a positively skewed distribution with mean 18.4 seconds and standard deviation 6.5 seconds. A random sample of 52 independent queries is logged.

(a) State the approximate sampling distribution of the sample mean waiting time $\bar{T}$, justifying your answer. **[2 marks]**

(b) Find the probability that the mean waiting time of these 52 queries exceeds 20.0 seconds. **[3 marks]**

2 A materials laboratory tests two properties of architectural glass panels:

- Refractive index R, which is known to be normally distributed with mean 1.520 and standard deviation 0.012.
- Impact resistance M J, which follows a heavily skewed distribution with mean 45.0 J and standard deviation 9.0 J.

A random sample of 16 glass panels is selected.

(a) Find the probability that the sample mean refractive index $\bar{R}$ of the 16 panels is less than 1.514. **[3 marks]**

(b) Explain why the normal calculation used in part (a) is valid for $\bar{R}$ when $n = 16$, whereas a normal calculation cannot be justified for the sample mean impact resistance $\bar{M}$ of the same 16 panels. **[2 marks]**

(c) If the sample size is increased to $n = 36$ panels, state the approximate distribution of $\bar{M}$. **[1 mark]**

3 The proportion of cloud cover, X, over a solar farm at midday is modelled as a continuous random variable with probability density function:

$$f(x) = \begin{cases} 3(1-x)^2 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show by integration that $\mathrm{E}(X) = \frac{1}{4}$ and $\mathrm{Var}(X) = \frac{3}{80}$. **[4 marks]**

(b) The midday cloud cover is recorded on 60 randomly chosen days, which may be assumed independent. Find the approximate probability that the sample mean cloud cover $\bar{X}$ over these 60 days is greater than 0.28. **[3 marks]**

4 The number of warranty claims, C, submitted per week by an appliance retailer has the discrete probability distribution shown below:

c	0	1	2	3
$\mathrm{P}(C = c)$	0.35	0.40	0.15	0.10

(a) Calculate $\mathrm{E}(C)$ and $\mathrm{Var}(C)$. **[3 marks]**

(b) For a random sample of 45 independent weeks, use the Central Limit Theorem to find the 95th percentile of the sample mean number of claims per week, $\bar{C}$. **[3 marks]**

5 ⋆ The mass of recycled aggregate in a dumper load, X tonnes, has an unknown distribution with mean 14.0 tonnes and standard deviation 2.4 tonnes. A site auditor takes a random sample of n independent loads, where $n \geq 30$.

(a) Find the minimum sample size n required so that the probability of the sample mean $\bar{X}$ differing from 14.0 tonnes by more than 0.5 tonnes is at most 0.05. **[4 marks]**

(b) Using $n = 89$, find the approximate probability that the total mass of the 89 loads exceeds 1270 tonnes. **[2 marks]**

6 Two automated sorting hubs, Hub North and Hub South, process parcels whose handling times follow non-normal distributions. At Hub North, handling time has mean 4.8 s and standard deviation 1.5 s. At Hub South, handling time has mean 4.5 s and standard deviation 1.2 s. Independent random samples of 50 parcels from Hub North and 40 parcels from Hub South are timed, with sample means $\bar{X}_N$ and $\bar{X}_S$ respectively.

(a) Justifying your method, state the approximate distribution of $\bar{X}_N - \bar{X}_S$. **[3 marks]**

(b) Find the approximate probability that the sample mean handling time at Hub South exceeds the sample mean handling time at Hub North. **[2 marks]**

4.4 Confidence Intervals for the Mean

A point estimate such as $\bar{x}$ gives one best estimate of an unknown population mean, but it does not show the uncertainty caused by sampling. A **confidence interval** adds this missing information by extending a margin of error on either side of the estimate:

$$\text{estimate} \ \pm \ (\text{critical value})(\text{standard error}).$$

The standard error measures how much the sample mean varies from sample to sample; the critical value determines how much of that sampling distribution the interval is designed to cover. The result is a range of plausible values for the fixed population mean μ, not a range containing a stated percentage of the observations.

4.4.1 Choosing between the z and t distributions

The formula you use to construct a confidence interval depends on what information is available about the population standard deviation (σ) and the size of your sample (n).

There are three rules for selecting the correct distribution:

- If the population standard deviation σ is **known**, use the Normal distribution (z-values) for a normal population; for a non-normal population, a sufficiently large sample is still needed for a Central Limit Theorem approximation.
- If σ is **unknown** and the sample size is sufficiently large for the setting, you can estimate σ using the sample standard deviation s and use the Normal approximation.
- If σ is **unknown** and the sample size is **small** ($n < 30$), use the Student's t-distribution with $n - 1$ degrees of freedom, provided the population is approximately normal (or the data give no serious evidence against that assumption).

Key Point

Confidence interval (σ known):

$$\bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Confidence interval (σ unknown, small n):

$$\bar{x} \pm t_{n-1}\left(\frac{s}{\sqrt{n}}\right)$$

Where $\bar{x}$ is the sample mean, n is the sample size, and s is the sample standard deviation.

Example 1

Question: A machine produces steel rods. The standard deviation of the length of the rods is known to be 4 mm. A random sample of 36 rods is taken, and the sample mean length is found to be 250 mm. Calculate a 95% confidence interval for the true mean length of the rods.

Solution: We know $\sigma = 4$, $n = 36$, and $\bar{x} = 250$. Because σ is known, we use the z-distribution. For a 95% confidence interval, the critical z-value is 1.96 (leaving 2.5% in each tail).

$$\text{CI} = \bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$\text{CI} = 250 \pm 1.96\left(\frac{4}{\sqrt{36}}\right)$$

$$\text{CI} = 250 \pm 1.96\left(\frac{4}{6}\right)$$

$$\text{CI} = 250 \pm 1.307$$

The 95% confidence interval is $(248.69 \text{ mm}, 251.31 \text{ mm})$.

Example 2

Question: A biologist is studying the weight of a rare species of frog. She catches a random sample of 10 frogs and records their weights. The sample mean is 12.5 g and the sample standard deviation is 2.1 g. Calculate a 99% confidence interval for the mean weight of this species of frog.

Solution: We have $n = 10$, $\bar{x} = 12.5$, and $s = 2.1$. Because σ is **unknown** and the sample size is **small** ($n < 30$), we **must** use the t-distribution.
Degrees of freedom, $\nu = n - 1 = 10 - 1 = 9$. For a 99% confidence interval, we look up the t-value for $\nu = 9$ with 0.005 in each tail. From the t-distribution tables, $t_{9,0.005} = 3.250$.

$$\text{CI} = \bar{x} \pm t_{n-1}\left(\frac{s}{\sqrt{n}}\right)$$

$$\text{CI} = 12.5 \pm 3.250\left(\frac{2.1}{\sqrt{10}}\right)$$

$$\text{CI} = 12.5 \pm 3.250(0.664)$$

$$\text{CI} = 12.5 \pm 2.158$$

The 99% confidence interval is $(10.34 \text{ g}, 14.66 \text{ g})$.

4.4.2 Interpreting Confidence Intervals

A very common exam pitfall is misinterpreting what a confidence interval actually means.

Common Misconception

Never state "there is a 95% probability that the population mean lies in this interval." The population mean is a fixed constant, not a random variable: it is either in the interval or it is not. The 95% refers to the *procedure* — if many samples were taken and an interval formed from each, about 95% of those intervals would contain μ.

Key Point

Correct Interpretation: "We are 95% confident that this interval captures the true population mean." This means that if we were to take 100 different samples and calculate 100 different confidence intervals, approximately 95 of them would contain the true population mean.

Exercise 4E

1 The breaking torque of ceramic bolts produced by a kiln is normally distributed with known population standard deviation $\sigma = 1.8$ N m. A random sample of 36 bolts has a sample mean breaking torque of 24.5 N m.

(a) Calculate a 95% confidence interval for the population mean breaking torque μ. **[3 marks]**

(b) The kiln specification states that the mean breaking torque is 25.2 N m. Use your confidence interval to comment on whether the kiln is meeting this specification. **[2 marks]**

2 An agronomist measures the nitrate concentration, x ppm, in a random sample of 12 soil extracts from an organic pasture. The sample data are summarised by:

$$\sum x = 318.0, \qquad \sum (x - \bar{x})^2 = 108.9$$

(a) Calculate unbiased point estimates of the population mean and the population variance. **[2 marks]**

(b) Calculate a 99% confidence interval for the population mean nitrate concentration. **[3 marks]**

(c) State the distributional assumption about the population of nitrate concentrations required for the interval in part (b) to be valid. **[1 mark]**

3 A biomedical engineer tests the flow rate, in L min^{-1}, of 8 randomly selected dialysis pumps of a new design. The recorded flow rates are:

$$0.48, \quad 0.52, \quad 0.51, \quad 0.47, \quad 0.50, \quad 0.53, \quad 0.49, \quad 0.50$$

Assume that flow rates in the population are normally distributed.

(a) Calculate the sample mean $\bar{x}$ and the unbiased sample standard deviation s. **[3 marks]**

(b) Construct a 90% confidence interval for the population mean flow rate μ. **[2 marks]**

(c) Two colleagues make the following statements about your interval:

- Colleague 1: "90% of all dialysis pumps of this design have a flow rate inside this interval."
- Colleague 2: "The population mean μ varies from sample to sample and lies inside this fixed interval 90% of the time."

Explain why both statements are statistically incorrect and write a correct interpretation of the 90% confidence level. **[2 marks]**

4 A broadband regulator monitors evening download speeds for rural households. Download speeds have a skewed distribution with unknown mean μ and unknown variance σ^2. A random sample of 80 rural households gives a sample mean of 38.4 Mbps and an unbiased sample standard deviation of $s = 11.2$ Mbps.

(a) Calculate a 98% confidence interval for the population mean evening download speed μ. **[3 marks]**

(b) Explain why the Central Limit Theorem is essential for the validity of this interval, and state what additional approximation is made because σ is unknown. **[2 marks]**

5 For each of the following independent sampling scenarios, state whether a confidence interval for the population mean μ should be constructed using the standard normal (z) distribution, Student's t-distribution (specifying the degrees of freedom), or whether neither method is justified from the information given. Give a brief reason for each decision.

(a) A random sample of $n = 14$ observations from a normal population with unknown variance σ^2. **[2 marks]**

(b) A random sample of $n = 18$ observations from a strongly skewed population with known standard deviation $\sigma = 5.2$. **[2 marks]**

(c) A random sample of $n = 15$ observations from a normal population with known standard deviation $\sigma = 2.4$. **[2 marks]**

6 $\star$ A metallurgist measures the Vickers hardness of a random sample of 10 titanium alloy specimens, assuming the hardness in the population is normally distributed. Using Student's t-distribution, the 95% confidence interval for the population mean hardness μ is reported as $(341.48, 354.52)$.

(a) Find the sample mean $\bar{x}$ and the unbiased sample standard deviation s. **[4 marks]**

(b) Using the same sample data, calculate a 99% confidence interval for μ. **[2 marks]**

4.4.3 Planning Sample Size and Interval Width

The **margin of error** is the amount added and subtracted from the sample mean. The total **width** of a confidence interval is exactly twice the margin of error.

Key Point

$$\text{Width of CI} = \text{Upper Limit} - \text{Lower Limit} = 2 \times \text{Margin of Error}$$

Because $\sqrt{n}$ is in the denominator of the margin of error, increasing the sample size n will decrease the margin of error, resulting in a narrower (more precise) confidence interval.

Example 3

Question: A machine fills bags with flour. The standard deviation of the weight is known to be 4 g. A quality control inspector wants to estimate the mean weight μ using a 95% confidence interval. Find the minimum sample size required to ensure the total width of the confidence interval is strictly less than 2 g.

Solution: The formula for the width of a z-interval is:

$$\text{Width} = 2 \times z \times \frac{\sigma}{\sqrt{n}}$$

For a 95% CI, $z = 1.96$. We require the width to be less than 2:

$$2 \times 1.96 \times \frac{4}{\sqrt{n}} < 2$$

$$\frac{15.68}{\sqrt{n}} < 2$$

Multiply both sides by $\sqrt{n}$ and divide by 2:

$$7.84 < \sqrt{n}$$

Square both sides:

$$n > 61.4656$$

Since the sample size n must be a whole number, the minimum sample size required is $n = 62$.

Exercise 4F

1 The viscosity of synthetic engine oil follows a normal distribution with known standard deviation $\sigma = 3.6$ cSt. A chemical engineer wishes to construct a 95% confidence interval for the population mean viscosity μ such that the total width of the interval is less than 1.5 cSt. Find the minimum sample size n required. **[4 marks]**

2 A supermarket chain audits the mass of pre-packed cheddar blocks. The population standard deviation of the block masses is known to be $\sigma = 6.0$ g.

(a) Find the minimum sample size required so that the auditor can be 99% confident that the sample mean is within 2.0 g of the true population mean. **[3 marks]**

(b) Suppose the auditor later wishes to reduce the maximum margin of error from 2.0 g to 0.5 g while keeping the confidence level at 99%. By what factor must the sample size be multiplied? Justify your answer algebraically. **[2 marks]**

3 The thickness of silicon wafers follows a normal distribution with known standard deviation $\sigma = 0.80\,\mu$m. A random sample of 64 wafers yields the confidence interval $(199.825, 200.175)\,\mu$m for the population mean thickness.

(a) Find the margin of error of the confidence interval and the standard error of the sample mean. **[2 marks]**

(b) Determine the percentage confidence level used to construct this interval, giving your answer to the nearest whole number. **[3 marks]**

4 ⋆ A hydrologist constructs confidence intervals for the mean flow velocity of a river, where the population standard deviation is σ m s^{-1}.

- Using a random sample of size n, a 90% confidence interval has total width 0.658 m s^{-1}.
- Increasing the sample size by 24 observations while keeping the confidence level at 90% reduces the total width of the interval to 0.470 m s^{-1}.

(a) Find the original sample size n. **[3 marks]**

(b) Hence find the value of σ. **[2 marks]**

5 A random sample $x_1, x_2, \ldots, x_n$ from a normal population with unknown variance is used to construct a 95% t-confidence interval for the population mean temperature μ (measured in °C). State, with a mathematical reason, how the width of the 95% confidence interval is affected by each of the following separate changes:

(a) Every temperature reading in the sample is converted from degrees Celsius (x) to kelvins ($y = x + 273.15$). **[2 marks]**

(b) Every temperature reading in the sample is converted from degrees Celsius (x) to degrees Fahrenheit ($f = 1.8x + 32$). **[2 marks]**

(c) A student mistakenly uses the normal critical value $z_{0.025} = 1.96$ instead of the correct $t_{n-1,0.025}$ critical value for the small sample. **[1 mark]**

6 A researcher takes a small pilot sample of $n = 5$ observations from a normal population, obtaining an unbiased sample standard deviation of $s = 4.0$, and constructs a 95% confidence interval for μ.

(a) Calculate the total width of this 95% confidence interval. **[2 marks]**

(b) A second independent sample of $n = 20$ observations from the same normal population also happens to have $s = 4.0$. Calculate the total width of the new 95% confidence interval, and explain why quadrupling n from 5 to 20 reduces the width of a t-interval by more than a factor of 2. **[3 marks]**

End-of-Chapter Mixed Exercises

1 A specialist tea company sells gift tins containing pouches of Assam tea and pouches of Darjeeling tea. The mass of an empty tin, T g, follows $\mathrm{N}(140, 4^2)$. The mass of a pouch of Assam tea, A g, follows $\mathrm{N}(125, 3^2)$, and the mass of a pouch of Darjeeling tea, D g, follows $\mathrm{N}(100, 2.5^2)$. All masses are independent.

(a) A standard gift tin consists of 1 tin, 2 pouches of Assam tea and 2 pouches of Darjeeling tea. Find the probability that the total mass of a standard gift tin exceeds 600 g. **[3 marks]**

(b) Find the probability that 4 randomly chosen pouches of Assam tea have a smaller combined mass than 5 randomly chosen pouches of Darjeeling tea. **[3 marks]**

2 An agricultural drone sprays liquid fertiliser onto orchard trees. The volume applied to a single tree, V litres, is normally distributed with mean 1.60 litres and standard deviation 0.15 litres.

(a) Find the mean and variance of the total volume applied to 9 independent trees, $S = V_1 + V_2 + \cdots + V_9$, and compare these with the mean and variance of $9V$. **[3 marks]**

(b) The drone's tank holds 15.3 litres. Find the probability that the tank contains enough fertiliser to spray 9 independent trees, and calculate the probability a student would mistakenly obtain if they modelled the total demand of the 9 trees as $9V$. **[3 marks]**

3 A civil engineer measures the compressive strength, x MPa, of 9 concrete cylinders cast from a new rapid-hardening mix. The results are summarised by:

$$\sum x = 378.0, \qquad \sum x^2 = 16002.0$$

(a) Calculate unbiased point estimates of the population mean μ and the population variance σ^2. **[3 marks]**

(b) Selecting an appropriate distribution and stating any necessary assumption about the population, calculate a 95% confidence interval for μ. **[4 marks]**

4 Let X_1, X_2, X_3, X_4 be a random sample of 4 independent observations from a population with mean μ and variance σ^2. Two statistics are proposed to estimate μ:

$$M_1 = k(X_1 + 2X_2 + 2X_3 + X_4), \qquad M_2 = \frac{X_1 + X_2 + X_3 + X_4}{4}$$

(a) Find the value of the constant k for which M_1 is an unbiased estimator of μ. **[2 marks]**

(b) Using your value of k, find $\mathrm{Var}(M_1)$ and $\mathrm{Var}(M_2)$ in terms of σ^2, and determine which estimator is more efficient. **[3 marks]**

(c) Find $\mathrm{E}(M_1 - M_2)$ and explain what this tells you about $M_1 - M_2$ as an estimator of μ. **[1 mark]**

5 $\star$ When a high-frequency sensor rounds timestamps, the rounding error E milliseconds is modelled by a continuous uniform distribution over the interval $[-3, 3]$, so that $\mathrm{E}(E) = 0$ and $\mathrm{Var}(E) = 3$ ms^2.

(a) A batch of 48 independent timestamps is processed. State the approximate distribution of the sample mean rounding error $\bar{E}$, justifying your answer, and find $\mathrm{P}(|\bar{E}| > 0.4)$. **[4 marks]**

(b) Find the minimum number of independent timestamps n that must be averaged so that $\mathrm{P}(|\bar{E}| < 0.2) \geq 0.95$. **[3 marks]**

6 A hospital pharmacy audits the waiting time, W minutes, for outpatient prescriptions to be dispensed. Waiting times are known to be positively skewed. In an initial audit of 64 randomly selected prescriptions, the sample mean waiting time is 21.5 minutes and the unbiased sample variance is $s^2 = 51.84$ minutes^2.

(a) Explain why it is valid to construct a normal (z) confidence interval for the population mean waiting time μ from this sample, and calculate a 95% confidence interval for μ. **[4 marks]**

(b) Treating $s = 7.2$ minutes as a reliable estimate of σ, find the minimum total sample size n required in a follow-up audit so that a 95% confidence interval for μ has a total width of at most 2.0 minutes. **[3 marks]**

7 An electric ferry crosses a strait in two phases: main crossing time X minutes and docking time Y minutes. From historical records, $X \sim \mathrm{N}(26.0, 2.4^2)$ and $Y \sim \mathrm{N}(5.5, 1.8^2)$, where X and Y are independent.

(a) Find the probability that a single complete journey ($J = X + Y$) takes longer than 35.0 minutes. **[3 marks]**

(b) Over a week, 16 independent complete journeys are monitored. Find the probability that the sample mean journey time $\bar{J}$ of these 16 journeys lies between 30.0 minutes and 32.7 minutes, stating with a reason whether the Central Limit Theorem is required. **[3 marks]**

(c) If the ferry completes an outbound journey J_1 and an immediate independent return journey J_2, find the probability that the total time $J_1 + J_2$ exceeds 70.0 minutes. **[2 marks]**

8 A battery manufacturer claims that its solid-state cells have a mean internal resistance of $\mu = 15.0$ mΩ. An independent laboratory tests a random sample of 6 cells and records the following internal resistances (in mΩ):

$$15.2, \quad 16.2, \quad 14.9, \quad 15.8, \quad 16.5, \quad 15.6$$

Internal resistance of these cells may be assumed to follow a normal distribution.

(a) Calculate the sample mean $\bar{x}$ and show that the unbiased sample variance is $s^2 = 0.36$ (mΩ)^2. **[3 marks]**

(b) Construct a 95% confidence interval for the population mean internal resistance μ, and state whether the interval supports the manufacturer's claim that $\mu = 15.0$ mΩ. **[3 marks]**

(c) Explain why a 99% confidence interval constructed from the same 6 cells does include 15.0 mΩ, and explain why this is not a contradiction. **[1 mark]**

9 A cloud-computing platform charges a research laboratory for nightly data-processing batch jobs. Each job requires a CPU processing time X hours and a GPU processing time Y hours, where $X \sim \mathrm{N}(6.85, 0.8^2)$ and $Y \sim \mathrm{N}(2.00, 0.5^2)$ independently. Each nightly job incurs a fixed base fee of £15, plus £4 per hour of CPU time and £12 per hour of GPU time, so that the total nightly cost, in pounds, is $C = 15 + 4X + 12Y$.

(a) Find the mean and standard deviation of the total nightly cost C. A technician calculates the variance of C as $15^2 + 4^2(0.8^2) + 12^2(0.5^2) = 271.24$. Identify the technician's error and explain why the fixed base fee does not affect the variance. **[3 marks]**

(b) Find the probability that the total cost of a single nightly batch job exceeds £76.60. **[2 marks]**

(c) Find the probability that, on a randomly chosen night, the GPU cost ($12Y$) exceeds the CPU cost ($4X$). **[3 marks]**

10 A marine acoustic engineer measures the signal attenuation, x dB/km, across 12 independent test transmissions in coastal water. To simplify recording, the engineer codes the measurements using $y = x - 18.0$ and obtains the summary statistics:

$$\sum y = 15.6, \qquad \sum y^2 = 41.84$$

(a) Calculate unbiased estimates of the population mean μ and the population variance σ^2 of the signal attenuation x, explaining briefly why subtracting 18.0 affects the mean but not the variance. **[4 marks]**

(b) Stating any necessary distributional assumption, calculate a 90% confidence interval for μ. **[3 marks]**

(c) The engineer writes in a report: *"There is a* 0.90 *probability that the true population mean* μ *lies between the two limits of this confidence interval."* Explain why this statement is statistically incorrect, and give a correct interpretation of the 90% confidence level. **[2 marks]**

11 A logistics depot loads courier parcels onto a light cargo aircraft. The mass of an individual parcel, M kg, follows a positively skewed continuous distribution with mean $\mu = 4.65$ kg and standard deviation $\sigma = 1.80$ kg. A consignment of 64 parcels is selected at random from the depot to be loaded into a cargo bay.

(a) State the value of $\mathrm{E}(\bar{M})$ and calculate the standard error of the sample mean mass $\bar{M}$. Explain why $\bar{M}$ is not *exactly* normally distributed, and state the theorem that justifies using a normal approximation for $\bar{M}$. **[3 marks]**

(b) The cargo bay has a maximum safe payload limit of 312.0 kg. Find the approximate probability that the total mass of the 64 parcels exceeds this safe limit. **[3 marks]**

(c) On a particular flight, all 64 parcels in the consignment come from a single bulk order of identical textbooks packed by one publisher. State, with a reason, how this affects the validity of your calculation in part (b). **[1 mark]**

12 In the manufacture of optical fibre connectors, a cylindrical ceramic ferrule of outer diameter X mm must fit inside a metal alignment sleeve of inner diameter Y mm. In routine production, X and Y are independent normal random variables with $X \sim \mathrm{N}(2.495, 0.003^2)$ and $Y \sim \mathrm{N}(\mu_Y, 0.004^2)$.

(a) Initially, the sleeve machine is calibrated so that $\mu_Y = 2.506$ mm. Find the probability that a randomly chosen ferrule cannot fit inside a randomly chosen sleeve (that is, $Y < X$). **[3 marks]**

(b) For optimal optical alignment, the diametral clearance $C = Y - X$ must lie between 0.002 mm and 0.018 mm. With $\mu_Y = 2.506$ mm, find the proportion of randomly paired connectors that meet this specification. **[3 marks]**

(c) Following a contract upgrade, the ferrule process and $\sigma_Y = 0.004$ mm remain unchanged, but the mean sleeve diameter μ_Y is recalibrated so that the probability of a ferrule failing to fit inside a sleeve ($Y < X$) is reduced to 0.001. Find the required value of μ_Y, giving your answer to 4 decimal places. **[3 marks]**

13 An environmental chemist measures the dissolved nitrate concentration, x mg/L, in water samples taken from an estuary.

(a) In Survey A, a large random sample of n water samples yields a sample mean $\bar{x}$ and an unbiased sample standard deviation $s = 1.60$ mg/L. Using these values, the chemist calculates a 95% confidence interval for the population mean nitrate concentration μ to be $(11.908, 12.692)$ mg/L. Find the value of $\bar{x}$, the standard error of the mean, and the sample size n. **[3 marks]**

(b) Using the same sample data from Survey A, a regulatory summary quotes the confidence interval $(11.835, 12.765)$ mg/L. Determine the percentage confidence level used for this second interval. **[3 marks]**

(c) In a separate winter survey (Survey B) at a tributary site, only 8 water samples are collected, giving $\bar{x} = 12.30$ mg/L and $s = 1.60$ mg/L. Without calculating the new interval, give two reasons why a 95% confidence interval from Survey B will be wider than $(11.908, 12.692)$ mg/L,

and state an additional distributional assumption required in Survey B that was not required in Survey A. **[3 marks]**

14 ⋆ A metrology institute estimates the true frequency μ of a reference oscillator using two independent measurement rigs. Rig 1 has measurement variance σ^2 and takes a random sample of 10 independent readings with sample mean $\bar{X}$. Rig 2 is an older instrument with measurement variance $2\sigma^2$ and takes a random sample of 5 independent readings with sample mean $\bar{Y}$. Both rigs produce unbiased readings, so that $\mathrm{E}(\bar{X}) = \mu$ and $\mathrm{E}(\bar{Y}) = \mu$. A physicist combines the two sample means using the estimator

$$T = w\bar{X} + (1 - w)\bar{Y}$$

where w is a real constant.

(a) Show that T is an unbiased estimator of μ for all values of w, and write down $\mathrm{Var}(\bar{X})$ and $\mathrm{Var}(\bar{Y})$ in terms of σ^2. **[2 marks]**

(b) Express $\mathrm{Var}(T)$ in terms of w and σ^2, and hence find the value of w for which T is the most efficient estimator of μ. **[4 marks]**

(c) A colleague suggests discarding the readings from Rig 2 and using $\bar{X}$ alone to estimate μ. Find the percentage reduction in variance achieved by using the optimal estimator T from part (b) instead of $\bar{X}$ alone. **[2 marks]**

15 A sports science laboratory investigates the aerodynamic drag reduction of a new cycling skinsuit compared with a standard jersey at 40 km/h in a wind tunnel. Across the population of club cyclists, the power required in the standard jersey is $X \sim \mathrm{N}(285, 12^2)$ watts and the power required in the new skinsuit is $Y \sim \mathrm{N}(270, 9^2)$ watts.

(a) One cyclist wearing the standard jersey and an independent cyclist wearing the new skinsuit are selected at random from the population. Find the probability that the cyclist in the new skinsuit requires more power than the cyclist in the standard jersey ($Y > X$). **[3 marks]**

(b) Explain why it would be invalid to use $\mathrm{Var}(X - Y) = 12^2 + 9^2 = 225$ W^2 to model the power saving $W = X - Y$ when the *same* cyclist is tested in both garments. **[1 mark]**

(c) In a paired trial, 10 randomly selected cyclists each test both garments, and the individual power saving $w = x - y$ watts is recorded for each cyclist. The results are summarised by:

$$\sum w = 150.0, \qquad \sum (w - \bar{w})^2 = 202.5$$

Assuming that individual power savings are normally distributed, calculate a 99% confidence interval for the population mean power saving μ_W. **[4 marks]**

(d) The manufacturer advertises that the skinsuit reduces power demand at 40 km/h by at least 12.0 W on average. State, with a reason, what conclusion can be drawn about this claim from your 99% confidence interval. **[1 mark]**

Chapter 4 Summary

Summary of Key Formulas and Concepts

- Expectations combine linearly; variances add only for independent variables and scale by squared constants.
- A multiple of one observation is not the same random variable as a sum of independent observations.
- For a large sample, the central limit theorem justifies an approximately normal sampling distribution of the mean.
- An interval estimate must be reported with its confidence level and interpreted as a repeated-sampling procedure.

Appendix A

Calculator Guide

CCEA expects candidates to use calculator technology confidently for summary statistics, probability distributions and inference. This appendix gives the actual key sequences for the three calculators most commonly used in UK classrooms:

- **Casio ClassWiz fx-991EX**;
- **Casio ClassWiz fx-991CW**;
- **Casio fx-CG50** graphical calculator.

Menu wording can differ slightly after an operating-system update. Practise these procedures on your own calculator before the examination.

Key Point

Examination rules. A calculator must not have computer algebra, internet access or unapproved stored files. Statistical output does not replace written working: state the model or hypotheses, show the statistic or probability used, and give a conclusion in context.

A.1 Generating Random Integers

To take a simple random sample, number the sampling units 1 to N, generate integers in that range, discard repeats when sampling without replacement, and continue until the sample is full.

Casio fx-991EX

1. Press **ALPHA**, then the decimal-point key `.` to enter `RanInt#(`.
2. Enter the lower bound. Press **SHIFT**, `)` for a comma, then enter the upper bound and press `)`.
3. For units 1 to 50, the display should read `RanInt#(1,50)`. Press = once for each integer required.

Casio fx-991CW

1. Press **CATALOG**, choose **Probability**, then **Random Integer** and **EXE**.
2. Enter the lower bound, a comma (**SHIFT**, `)`), and the upper bound; close the bracket.
3. Press **EXE** repeatedly to generate further integers.

Casio fx-CG50

1. From **Run-Matrix**, press **OPTN**, **F6** (▷), **F3** (PROB), **F4** (RAND), **F2** (Int).
2. Enter the lower bound, the dedicated comma key, and the upper bound; close the bracket.
3. Press **EXE** once for each integer required.

Record the sampling frame, numbering convention, range, treatment of repeats and final sample.

A.2 One-Variable Statistics

The one-variable results include n, $\bar{x}$, $\sum x$, $\sum x^2$, population standard deviation σ_x and sample standard deviation s_x.

Key Point

Use s_x (division by $n-1$) for the unbiased sample variance or standard deviation used in estimation and t-tests. Use σ_x only when the entered values are the complete population. Square the appropriate standard deviation when a variance is required.

Casio fx-991EX

1. For a frequency table, press **SHIFT**, **MENU** (SETUP), ↓, **3: Statistics**, **1: On**. For raw data, select **2: Off**.
2. Press **MENU**, **6: Statistics**, **1: 1-Variable**.
3. Enter each value in the x column, pressing = after each entry. If frequency is on, enter the corresponding counts in the Freq column.
4. Press **OPTN**, **3: 1-Variable Calc**. Scroll to read n, $\bar{x}$, $\sum x$, $\sum x^2$, σ_x and s_x.

Casio fx-991CW

1. Press **HOME**, select **Statistics**, then **1-Variable**.
2. If frequencies are needed, press **TOOLS**, **Frequency**, **On**, then the curved **Back** key.
3. Enter the values and, where applicable, their frequencies. Press **EXE** after the last entry.
4. Choose **1-Var Results** and scroll through the output.

Casio fx-CG50

1. Press **MENU**, **2: Statistics**. Put values in **List 1** and optional frequencies in **List 2**.
2. Press **F2** (CALC), **F6** (SET). Set `1Var XList` to **List1**. Set `1Var Freq` to **List2** for grouped frequencies or **1** for raw data.
3. Press **EXIT**, then **F1** (1-VAR). Scroll through the results.

A.3 Pearson Correlation and Linear Regression

Enter the explanatory variable as x and the response as y. Choose $y = a + bx$: a is the intercept, b the gradient, and r Pearson's product-moment correlation coefficient.

Casio fx-991EX

1. Press **MENU**, **6: Statistics**, **2:** $y = a + bx$.
2. Enter the paired x- and y-values in their columns.
3. Press **OPTN**, **4: Regression Calc**; read a, b and r.

Casio fx-991CW

1. Press **HOME**, **Statistics**, **2-Variable**.
2. Enter the paired values and press **EXE** after the final entry.
3. Select **Reg Results** to obtain a, b and r. **2-Var Results** instead gives the separate summary statistics for the two columns.

Casio fx-CG50

1. Press **MENU**, **2: Statistics**; enter x in **List 1** and y in **List 2**.
2. Press **F2** (CALC), **F3** (REG), **F1** (X), **F2** ($a + bx$).
3. Read a, b and r. Check that you have not accidentally selected the regression of x on y.

A.4 Probability Distributions

Key Point

For ClassWiz discrete cumulative functions, `CD` means $P(X \leq x)$. Translate first:

$$P(X < k) = P(X \leq k-1), \qquad P(X \geq k) = 1 - P(X \leq k-1), \qquad P(X > k) = 1 - P(X \leq k).$$

Key Point

If $X \sim \mathrm{N}(\mu, \sigma^2)$, the calculator asks for the *standard deviation* σ, not the variance. For example, enter $\sigma = 4$ for $X \sim \mathrm{N}(50, 16)$.

Casio fx-991EX

Press **MENU**, **7: Distribution**, then select the required command. Choose **2: Variable** when asked how the data will be entered.

- **Normal CD:** enter Lower, Upper, σ, μ, pressing **=** after each. For a one-sided tail use a sufficiently extreme bound, such as -10^{99} or 10^{99}.
- **Inverse Normal:** enter the area to the *left*, then σ and μ.
- **Binomial PD:** enter x, Numtrial n, and p for $P(X = x)$.

- **Binomial CD:** enter x, Numtrial n, and p for $P(X \le x)$.
- **Poisson PD/CD:** enter x and the interval mean λ (shown as μ).

Casio fx-991CW

Press **HOME**, select **Distribution**, choose the distribution command and select **Variable**.

- **Normal CD:** enter Lower, Upper, σ, μ; press **EXE**.
- **Inverse Normal:** enter the left-tail Area, σ, μ.
- **Binomial PD/CD:** enter x, n, p.
- **Poisson PD/CD:** enter x and the interval mean λ.

Casio fx-CG50

Press **MENU**, **2: Statistics**, **F5** (DIST), and set `Data` to **Variable**.

- **Normal CD: F1** (NORM), **F2** (Ncd); enter Lower, Upper, σ, μ.
- **Inverse Normal: F1** (NORM), **F3** (InvN); choose Left, Right or Central tail and enter the area, σ, μ.
- **Binomial: F5** (BINM), then **F1** (Bpd) or **F2** (Bcd). For Bpd enter x, Numtrial and p; for Bcd enter Lower, Upper, Numtrial and p.
- **Poisson: F6** (▷), **F1** (POISN), then **F1** (Ppd) or **F2** (Pcd). Enter x or the bounds and the interval mean $\mu = \lambda$.

Geometric probabilities

These models do not require a distribution menu. In Calculate or Run-Matrix mode enter

$$P(X = x) = (1-p)^{x-1}p, \qquad P(X > x) = (1-p)^{x}.$$

For example, with $p = 0.2$, enter `0.8^(6-1)×0.2` to find $P(X = 6)$.

Continue in the full book

This evaluation copy includes Sections A.1–A.4 only. The complete book contains the rest of the calculator guide, including workflows for sample means, confidence intervals, t-tests and χ^2 tests.

A2 Statistics Practice Set 1

CCEA Further Mathematics Statistics

A2 Statistics Practice Set 1

Level: A2
Total marks: 75

Instructions

- Answer all questions in this Statistics section.
- A calculator may be used. Show sufficient working to make your method clear.
- Give numerical answers to three significant figures unless otherwise stated.
- The marks for each question or part are shown in brackets.

For practice use. Consult current CCEA materials for official assessment guidance.

1. A manufacturer produces two independent types of battery. The operating lives, in hours, satisfy $X \sim N(12, 4)$ and $Y \sim N(9, 9)$.

   (i) Find the distribution of $2X - Y$ and hence find $P(2X - Y > 22)$. **[6 marks]**

   (ii) Ten batteries of type X are selected independently. Find the distribution of their total operating life. **[5 marks]**

   (iii) Compare the variance in part (ii) with the variance of $10X$, where X represents one battery, and explain the difference. **[4 marks]**

2. The mass of material placed in a container has population mean 46 g and standard deviation 12 g.

   (i) For a random sample of 64 containers, state the approximate distribution of the sample mean and find $P(\bar{X} > 49)$. State why the approximation may be used. **[5 marks]**

   (ii) A different random sample of 36 containers gives $\bar{x} = 47.2$ g and $s = 11.2$ g. Using $t_{35,0.025} = 2.030$, calculate a 95% confidence interval for the population mean. **[6 marks]**

   (iii) Explain the repeated-sampling meaning of the phrase "95% confidence interval". **[4 marks]**

3. A laboratory claims that the mean concentration of an active ingredient is 50 mg. A random sample from a Normal population gives $n = 10$, $\bar{x} = 52.4$ and $s = 3.2$.

   (i) State hypotheses for testing whether the mean exceeds 50 mg, and state the condition needed for this test. **[4 marks]**

   (ii) Using $t_{9,0.05} = 1.833$, carry out the test at the 5% significance level. **[6 marks]**

   (iii) Using $t_{9,0.025} = 2.262$, calculate a 95% confidence interval for the population mean. **[5 marks]**

4. A trial records treatment outcome in the following 2×2 table.

	Success	Not successful
Treatment	32	18
Control	20	30

   (i) Calculate the four expected frequencies under independence. **[5 marks]**

   (ii) Calculate Yates' corrected χ^2 statistic. **[6 marks]**

   (iii) Test independence at the 5% level using the critical value 3.841, and explain why Yates' correction is used. **[4 marks]**

5. Six participants complete the same task before and after training. Their improvements, in seconds, are

$$4, \quad 2, \quad 5, \quad 1, \quad 3, \quad 4.$$

   (i) Explain why a paired test is appropriate and state hypotheses for a positive mean improvement. **[4 marks]**

   (ii) Calculate the sample mean and sample standard deviation of the improvements. **[5 marks]**

   (iii) Using $t_{5,0.05} = 2.015$, carry out the test at the 5% level and state the distributional condition required. **[6 marks]**

END OF SECTION
TOTAL FOR SECTION IS 75 MARKS

Answers

Linear Combinations, Sampling and Estimation

Exercise 4A

1. (a) 27
   (b) 53
   (c) 149

2. (a) Mean: 22.2 mm, Variance: 0.1092 mm^2
   (b) Mean: 0.6 mm, Standard deviation: $\sqrt{0.0481}$ mm ≈ 0.219 mm

3. (a) $\text{E}(B_1 + B_2 + B_3) = 540$ Wh, $\text{Var}(B_1 + B_2 + B_3) = 48$ Wh^2; $\text{E}(3B) = 540$ Wh, $\text{Var}(3B) = 144$ Wh^2. Both have the same mean, but $\text{Var}(3B)$ is three times $\text{Var}(B_1 + B_2 + B_3)$.
   (b) $3B$ represents taking a single randomly chosen battery pack and tripling its capacity (or using three identical copies of the same pack). $B_1 + B_2 + B_3$ has a smaller variance because random fluctuations above and below the mean across three independent packs partially cancel out, whereas any deviation in a single pack is tripled in $3B$.

4. (a) Mean: £3270, Standard deviation: $£\sqrt{274500} \approx £524$ (or £523.93)
   (b) Cars and vans compete for the same finite deck space on the ferry (yielding negative correlation when near capacity), and both are influenced by common factors such as weather or time of year (yielding positive correlation). Only the variance and standard deviation calculation relies on the independence assumption; the mean does not.

5. (a) $\text{E}(W) = k\,\text{E}(P) - 2\,\text{E}(Q) = 6k - 18 = 12 \implies 6k = 30 \implies k = 5$
   (b) $\text{Var}(W) = 5^2\,\text{Var}(P) + (-2)^2\,\text{Var}(Q) = 25\sigma^2 + 8\sigma^2 = 33\sigma^2 = (11\sqrt{3})^2 = 363 \implies \sigma^2 = 11$
   (c) The coefficient of P_3 must be squared when finding the variance of a linear combination of independent random variables: $\text{Var}(P_1 + P_2 - P_3) = \text{Var}(P_1) + \text{Var}(P_2) + (-1)^2\,\text{Var}(P_3) = 3\sigma^2 = 33$.

6. (a) $\text{E}(C) = w\,\text{E}(M_1) + (1 - w)\,\text{E}(M_2) = w\mu + (1 - w)\mu = \mu$
   (b) Since M_1 and M_2 are independent, $\text{Var}(C) = w^2\,\text{Var}(M_1) + (1 - w)^2\,\text{Var}(M_2) = 9w^2 + 16(1 - 2w + w^2) = (25w^2 - 32w + 16)$ mg^2L^{-2}.
   (c) $w = \frac{16}{25} = 0.64$, Minimum variance: $\frac{144}{25}$ $\text{mg}^2\text{L}^{-2} = 5.76$ mg^2L^{-2}

Exercise 4B

1. (a) $X + Y \sim \text{N}(52, 100)$, $\text{P}(X + Y < 60) = 0.788$
   (b) $2X - 3Y \sim \text{N}(4, 720)$, $\text{P}(2X - 3Y > 15) = 0.341$

2. (a) Mean = 72 kg, Variance = 1.834 kg^2
   (b) 0.0134

3. (a) $V_1 + V_2 + V_3 + V_4 \sim \text{N}(100, 1.44)$, 0.789
   (b) $4V \sim \text{N}(100, 5.76)$, $\text{P}(98.5 < 4V < 101.5) = 0.468$. The probability is smaller because $\text{Var}(4V) = 16\text{Var}(V) = 5.76$ ml^2 is greater than $\text{Var}(V_1+V_2+V_3+V_4) = 4\text{Var}(V) = 1.44$ ml^2; in Method A independent random errors partially cancel out, whereas in Method B any error in the single cycle is multiplied by 4.

4. (a) $P - Q \sim \text{N}(-0.8, 4)$, 0.655
   (b) 0.274

5. (a) $S - D \sim \text{N}(0.25, 0.0225)$, 0.794
   (b) $(D_1 + D_2 + D_3) - (S_1 + S_2 + S_3) \sim \text{N}(-0.75, 0.0675)$, 0.00195

6. (a) 35
   (b) $A + B \sim \text{N}(275, 25 + \sigma^2)$, $\frac{288-275}{\sqrt{25+\sigma^2}} = 1.96$, $\sigma^2 = 18.99$

Exercise 4C

1. (a) Parameter, because μ is a numerical characteristic of the entire population (all positions in the concert hall), not calculated from a sample.
   (b) $\sum x = 12.0$, $\sum x^2 = 24.26$; $\hat{\mu} = \bar{x} = \frac{12.0}{6} = 2.0$ s and $s^2 = \frac{1}{5}\left(24.26 - \frac{12.0^2}{6}\right) = \frac{0.26}{5} = 0.052$ s^2

(c) $\mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}} = \sqrt{\frac{0.052}{6}} = \frac{\sqrt{195}}{150} \approx 0.0931$ s

2. (a) $\bar{y} = \frac{136.0}{16} = 8.5$ mg L^{-1}; $\sum(y-\bar{y})^2 = 1174.0 - \frac{136.0^2}{16} = 18.0$, so $s^2 = \frac{18.0}{15} = 1.2$ (mg L^{-1})^2

(b) Sample observations are on average closer to the sample mean $\bar{Y}$ than to the population mean μ, so $\mathrm{E}\left[\frac{1}{n}\sum(Y-\bar{Y})^2\right] = \frac{n-1}{n}\sigma^2 < \sigma^2$, which systematically underestimates σ^2. Replacing μ by $\bar{y}$ imposes the single linear constraint $\sum(y-\bar{y}) = 0$, leaving $n-1$ independent deviations (degrees of freedom) free to vary.

3. (a) $\mathrm{E}(T_1) = \frac{\mu+3\mu+\mu}{5} = \mu$ and $\mathrm{E}(T_2) = \frac{2\mu+\mu}{3} = \mu$. Since $\mathrm{E}(T_1) = \mathrm{E}(T_2) = \mu$, both T_1 and T_2 are unbiased estimators of μ.

(b) $\mathrm{Var}(T_1) = \frac{1^2+3^2+1^2}{5^2}\sigma^2 = \frac{11}{25}\sigma^2 = 0.44\sigma^2$; $\mathrm{Var}(T_2) = \frac{2^2+1^2}{3^2}\sigma^2 = \frac{5}{9}\sigma^2 \approx 0.556\sigma^2$. T_1 is more efficient because $\mathrm{Var}(T_1) < \mathrm{Var}(T_2)$.

(c) $\bar{X}$ is also unbiased ($\mathrm{E}(\bar{X}) = \mu$) and weights all three independent observations equally, achieving a smaller variance $\mathrm{Var}(\bar{X}) = \frac{1}{3}\sigma^2$ than both T_1 and T_2.

4. (a) $\mathrm{E}(W) = \left(a+b+\frac{1}{2}\right)\mu = \mu \implies a+b+\frac{1}{2} = 1 \implies b = \frac{1}{2} - a$

(b) $\mathrm{Var}(W) = \left(a^2+b^2+\frac{1}{4}\right)\sigma^2 = \frac{3}{8}\sigma^2 \implies a^2 + \left(\frac{1}{2}-a\right)^2 = \frac{1}{8} \implies (4a-1)^2 = 0$, so $a = \frac{1}{4}$ and $b = \frac{1}{4}$

5. (a) $\mathrm{SE}(\bar{X}) = \frac{36}{\sqrt{18}} = 6\sqrt{2} \approx 8.49$ MPa; $\frac{36}{\sqrt{n}} = 3 \implies \sqrt{n} = 12 \implies n = 144$

(b) 1. The population standard deviation σ is a fixed parameter of the population and does not decrease when n increases (a larger sample reduces the standard error $\frac{\sigma}{\sqrt{n}}$, not σ). 2. Because the standard error is $\frac{\sigma}{\sqrt{n}}$, tripling the sample size divides the standard error by $\sqrt{3}$, not 3 (the sample size must be multiplied by 9 to divide the standard error by 3).

6. (a) Both samples have the same mean (6.4), so the combined mean is $\bar{x} = \frac{5(6.4)+6(6.4)}{5+6} = 6.4$.

(b) $\sum_A(x-6.4)^2 = 4(0.15) = 0.60$ and $\sum_B(x-6.4)^2 = 5(0.25) = 1.25$, so $\sum_{\text{all}}(x-6.4)^2 = 0.60 + 1.25 = 1.85$; $s^2 = \frac{1.85}{11-1} = 0.185$; estimated standard error $= \frac{s}{\sqrt{11}} = \sqrt{\frac{0.185}{11}} = \frac{\sqrt{814}}{220} \approx 0.130$

Exercise 4D

1. (a) $\bar{T} \sim \mathrm{N}(18.4, 0.8125)$ (or $\mathrm{N}(18.4, 0.901^2)$) approximately, by the Central Limit Theorem, since the sample of $n = 52$ queries is large and independent.

(b) $z = \frac{20.0-18.4}{6.5/\sqrt{52}} = 1.775$, $\mathrm{P}(\bar{T} > 20.0) = 0.0379$ (or 0.0375 from tables)

2. For $n = 16$:

- $\bar{R} \sim \mathrm{N}(1.520, 0.003^2)$
- $\mathrm{E}(\bar{M}) = 45.0$ J, $\mathrm{SD}(\bar{M}) = 2.25$ J

(a) $z = \frac{1.514-1.520}{0.012/\sqrt{16}} = -2.00$, $\mathrm{P}(\bar{R} < 1.514) = 0.0228$

(b) Refractive index R is normally distributed in the population, so $\bar{R}$ is exactly normally distributed for any sample size. Impact resistance M has a heavily skewed distribution and $n = 16$ is too small for the Central Limit Theorem to justify a normal approximation for $\bar{M}$.

(c) $\bar{M} \sim \mathrm{N}(45.0, 2.25)$ (or $\mathrm{N}(45.0, 1.5^2)$) approximately.

3. (a) $\mathrm{E}(X) = \int_0^1 3x(1-x)^2\,\mathrm{d}x = \int_0^1 (3x - 6x^2 + 3x^3)\,\mathrm{d}x = \left[\frac{3}{2}x^2 - 2x^3 + \frac{3}{4}x^4\right]_0^1 = \frac{1}{4}$. $\mathrm{E}(X^2) = \int_0^1 3x^2(1-x)^2\,\mathrm{d}x = \int_0^1 (3x^2 - 6x^3 + 3x^4)\,\mathrm{d}x = \left[x^3 - \frac{3}{2}x^4 + \frac{3}{5}x^5\right]_0^1 = \frac{1}{10}$. $\mathrm{Var}(X) = \frac{1}{10} - \left(\frac{1}{4}\right)^2 = \frac{3}{80}$.

(b) $\bar{X} \sim \mathrm{N}(0.25, 0.000625)$ approximately; $z = \frac{0.28-0.25}{0.025} = 1.20$, $\mathrm{P}(\bar{X} > 0.28) = 0.115$

4. (a) $\mathrm{E}(C) = 0(0.35) + 1(0.40) + 2(0.15) + 3(0.10) = 1$, $\mathrm{E}(C^2) = 0.40 + 0.60 + 0.90 = 1.9$, $\mathrm{Var}(C) = 1.9 - 1^2 = 0.9$

(b) $\bar{C} \sim \mathrm{N}(1, 0.02)$ approximately; $1 + 1.6449\sqrt{0.02} = 1.23$

5. (a) $\frac{0.5}{2.4/\sqrt{n}} \geq 1.96 \implies \sqrt{n} \geq 9.408 \implies n \geq 88.51$, so $n = 89$

   (b) $\sum X_i \sim \mathrm{N}(1246, 512.64)$ (or $\bar{X} \sim \mathrm{N}\left(14.0, \frac{5.76}{89}\right)$) approximately; $z = \frac{1270-1246}{2.4\sqrt{89}} = \frac{10}{\sqrt{89}} = 1.06$, P(total mass $>$ 1270) $= 0.145$

6. (a) Since both sample sizes ($n_N = 50$ and $n_S = 40$) are large, by the Central Limit Theorem $\bar{X}_N \sim \mathrm{N}(4.8, 0.045)$ and $\bar{X}_S \sim \mathrm{N}(4.5, 0.036)$ approximately. As the samples are independent, $\bar{X}_N - \bar{X}_S \sim \mathrm{N}(4.8-4.5, 0.045+0.036) = \mathrm{N}(0.3, 0.081)$ approximately.

   (b) $\mathrm{P}(\bar{X}_S > \bar{X}_N) = \mathrm{P}(\bar{X}_N - \bar{X}_S < 0)$; $z = \frac{0-0.3}{\sqrt{0.081}} = -1.054$, probability $= 0.146$ (or 0.147 from tables)

Exercise 4E

1. (a) $z = 1.96$, SE $= \frac{1.8}{\sqrt{36}} = 0.3$; $(23.91, 25.09)$ N m (or $(23.9, 25.1)$ N m)

   (b) Since 25.2 N m lies outside the 95% confidence interval, there is evidence that the kiln is not meeting the specification (the true mean breaking torque appears to be below 25.2 N m).

2. (a) $\bar{x} = 26.5$ ppm, $s^2 = 9.9$ ppm^2

   (b) $t_{11} = 3.106$, SE $= \sqrt{\frac{9.9}{12}} = 0.9083$; $(23.68, 29.32)$ ppm (or $(23.7, 29.3)$ ppm)

   (c) The nitrate concentrations in the population must be normally distributed.

3. (a) $\bar{x} = 0.50$ L min^{-1}, $s = 0.02$ L min^{-1}

   (b) $t_7 = 1.895$, SE $= \frac{0.02}{\sqrt{8}} = 0.007071$; $(0.487, 0.513)$ L min^{-1}

   (c) Colleague 1 is incorrect because the confidence interval is for the population mean μ, not individual pump flow rates. Colleague 2 is incorrect because μ is a fixed parameter, whereas the interval varies from sample to sample. Correct interpretation: under repeated random sampling of size 8, approximately 90% of the resulting confidence intervals would contain the true population mean flow rate μ.

4. (a) $z = 2.326$, SE $= \frac{11.2}{\sqrt{80}} = 1.252$; $(35.49, 41.31)$ Mbps (or $(35.5, 41.3)$ Mbps)

   (b) Because the population of download speeds is skewed, the Central Limit Theorem is required to ensure that the sample mean $\bar{X}$ is approximately normally distributed for a large sample ($n = 80$). Since σ is unknown, it is approximated by the unbiased sample standard deviation $s = 11.2$ Mbps.

5. (a) Student's t-distribution with 13 degrees of freedom, because the population is normal and σ^2 is unknown with a small sample size ($n = 14$).

   (b) Neither method is justified, because the population is strongly skewed and the sample size ($n = 18$) is too small for the Central Limit Theorem to apply.

   (c) Standard normal (z) distribution, because the population is normally distributed and the population standard deviation σ is known.

6. (a) $\bar{x} = 348$; using $t_9 = 2.262$ and margin of error 6.52, $s = 9.11$

   (b) Using $t_9 = 3.250$, $(338.63, 357.37)$ (or $(339, 357)$)

Exercise 4F

1. 89 ($2 \times 1.96 \times \frac{3.6}{\sqrt{n}} < 1.5 \implies n > 88.51$)

2. (a) 60 ($2.5758 \times \frac{6.0}{\sqrt{n}} \leq 2.0 \implies n \geq 59.71$)

   (b) Multiply by 16. Since $E = \frac{z\sigma}{\sqrt{n}}$, we have $n = \left(\frac{z\sigma}{E}\right)^2 \propto \frac{1}{E^2}$; dividing E by 4 (from 2.0 g to 0.5 g) multiplies n by $4^2 = 16$.

3. (a) Margin of error $= 0.175\,\mu$m, standard error $= 0.10\,\mu$m

   (b) 92% ($z = \frac{0.175}{0.10} = 1.75$, $\mathrm{P}(-1.75 < Z < 1.75) = 0.9199$)

4. - For sample size n: $\frac{2z_{0.05}\sigma}{\sqrt{n}} = 0.658$
   - For sample size $n + 24$: $\frac{2z_{0.05}\sigma}{\sqrt{n+24}} = 0.470$

(a) 25 ($\sqrt{\frac{n+24}{n}} = \frac{0.658}{0.470} = \frac{7}{5}$)

(b) 1.00 m s^{-1} (using $z_{0.05} = 1.645$)

5. (a) Unchanged, because adding a constant (273.15) shifts the sample mean but leaves the sample standard deviation s unchanged ($s_y = s_x$), so $2t_{n-1,0.025}\frac{s}{\sqrt{n}}$ is unchanged.

   (b) Multiplied by 1.8, because scaling by 1.8 and adding 32 multiplies the sample standard deviation by 1.8 ($s_f = 1.8 s_x$), which scales the width $2t_{n-1,0.025}\frac{s}{\sqrt{n}}$ by 1.8.

   (c) Decreased (narrower), because $z_{0.025} = 1.96 < t_{n-1,0.025}$ for any finite sample size n.

6. (a) 9.93 ($t_{4,0.025} = 2.776$, width $= 2 \times 2.7764 \times \frac{4.0}{\sqrt{5}}$)

   (b) 3.74 ($t_{19,0.025} = 2.093$, width $= 2 \times 2.0930 \times \frac{4.0}{\sqrt{20}}$). Quadrupling n from 5 to 20 halves the standard error $\frac{s}{\sqrt{n}}$ and simultaneously increases the degrees of freedom from 4 to 19, which decreases the critical value $t_{\nu,0.025}$ from 2.776 to 2.093; together these reduce the width by more than a factor of 2.

A2 Statistics Practice Set 1 Solutions

1. (i) $2X - Y \sim N(15, 25)$. Hence $P(2X - Y > 22) = P(Z > 1.4) = 0.0808$.

   (ii) $\sum X_i \sim N(120, 40)$.

   (iii) $10X \sim N(120, 400)$: multiplying one observation creates variance $10^2(4)$, whereas ten independent variances add to $10(4)$.

2. (i) By the Central Limit Theorem, since $n = 64$ is large, $\bar{X} \approx N(46, 12^2/64) = N(46, 2.25)$; $P(\bar{X} > 49) = P(Z > 2) = 0.0228$.

   (ii) $47.2 \pm 2.030(11.2/6)$ gives $(43.41, 50.99)$ g.

   (iii) In repeated sampling, about 95% of intervals constructed by this method contain the fixed population mean.

3. (i) $H_0 : \mu = 50$, $H_1 : \mu > 50$; the population is Normal.

   (ii) $t = (52.4 - 50)/(3.2/\sqrt{10}) = 2.372 > 1.833$; reject H_0. There is evidence that the mean concentration exceeds 50 mg.

   (iii) $52.4 \pm 2.262(3.2/\sqrt{10}) = (50.11, 54.69)$ mg.

4. (i) Row totals are $50, 50$ and column totals are $52, 48$, giving expected table $(26, 24; 26, 24)$.

   (ii) With Yates' correction, $\chi^2_Y = 2(5.5^2/26 + 5.5^2/24) = 4.848$.

   (iii) Since $4.848 > 3.841$, reject independence. There is evidence of an association between treatment and outcome. Yates' correction is a continuity correction for a 2×2 table with one degree of freedom.

5. (i) Measurements are linked within participants; $H_0 : \mu_d = 0$, $H_1 : \mu_d > 0$.

   (ii) $\bar{d} = 19/6 = 3.167$, $s_d = 1.472$.

   (iii) $t = 3.167/(1.472/\sqrt{6}) = 5.27 > 2.015$; reject H_0. There is evidence of positive mean improvement. The differences must be a random sample from a Normal population.